

Botnet Boasting: Investigating DDoS Power Proofs

Maarten Weyns
m.b.m.weyns@tudelft.nl
Delft University of Technology
Delft, Netherlands

Georgios Smaragdakis
g.smaragdakis@tudelft.nl
Delft University of Technology
Delft, Netherlands

Harm Griffioen
h.j.griffioen@tudelft.nl
Delft University of Technology
Delft, Netherlands

Abstract

Within the DDoS ecosystem, DDoS-for-Hire services have become increasingly popular, allowing anyone to launch DDoS attacks with the click of a button. To showcase their capabilities, these services often rely on third-party platforms that provide “DDoS Power Proofs”, where users can see real-time graphs of the traffic generated during an attack. We show that these platforms, known as DStat services, aggregate data from publicly exposed monitoring endpoints to display the current bandwidth usage and uplink capacities of DDoS providers. In this paper, we investigate the infrastructure behind DStat services, analyze the attacks performed on their endpoints, and explore the market dynamics of DDoS Power Proofs on Telegram. Our findings reveal that DStat services abuse publicly exposed monitoring endpoints, allowing them to operate without significant investment in infrastructure. We observe a wide range of attack sizes and characteristics, shedding light on the capabilities of the DDoS ecosystem as a whole. Furthermore, our analysis of Telegram groups uncovers the social dynamics and promotional strategies employed by DDoS providers.

CCS Concepts

• Security and privacy → Denial-of-service attacks.

Keywords

DDoS, DDoS-for-Hire, Botnet, Telegram

ACM Reference Format:

Maarten Weyns, Georgios Smaragdakis, and Harm Griffioen. 2026. Botnet Boasting: Investigating DDoS Power Proofs. In *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26)*, November 15–19, 2026, The Hague, Netherlands. ACM, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3830454.3846638>

1 Introduction

Distributed Denial of Service (DDoS) attacks continue to pose significant threats to online services, despite advancements in mitigation techniques. The ongoing evolution of attack methods and the corresponding development of defense mechanisms create a dynamic landscape where both attackers and defenders are continually adapting their strategies. This cat-and-mouse game results in increasingly sophisticated DDoS attacks that can bypass traditional mitigation systems. Recent trends show that attackers are increasingly capable of reaching Terabit-per-second (Tbps) levels of traffic [2, 8, 37].

In a DDoS attack, an attacker aims to exhaust resources on a victim system by sending a lot of network traffic from many locations around the world [24]. Modern DDoS attacks often leverage large botnets, which are networks of compromised devices that can be orchestrated to flood target systems with traffic. It is however not trivial to build and maintain such botnets. For this reason, attackers rent botnets from DDoS-for-Hire service providers (also called booters or stressers), which allow easy access to powerful attack capabilities for a fee [31, 32]. These service providers lower the barrier to entry for launching DDoS attacks, making them accessible to a wider range of malicious actors. As a result, the frequency and scale of DDoS attacks have increased significantly [2, 4, 34].

Around this DDoS-for-Hire ecosystem, a market has emerged where many different operators offer their attack services. These operators often compete on price, reliability, and the scale of attacks they can facilitate. From a user perspective, selecting a DDoS-for-Hire service can be challenging due to the lack of transparency, and the unknown firepower of the underlying botnet. Users may struggle to assess the effectiveness of different services, leading to uncertainty about the potential impact of a DDoS attack.

To address this problem of lack of trust, the underground economy has seen the emergence of DDoS testing services. These services allow users to identify the strength of a DDoS-for-Hire service before committing to a purchase, and are often shared in Telegram groups. These services typically offer a public dashboard where operators can showcase the extent of their DDoS capabilities by attacking what is essentially a trusted third party. This allows potential customers to evaluate the effectiveness of the DDoS-for-Hire service. These testing services are often referred to as “DStat” services, after the Linux monitoring tool `dstat` [3], which is commonly used to display real-time system resource usage statistics.

In this paper, we present a comprehensive study of the DStat ecosystem and the Telegram community backing it. We analyze DStat panel features, their operation, and the infrastructure being attacked. By understanding these dynamics, we can better comprehend the challenges defenders face in mitigating DDoS attacks.

The contributions of this paper are as follows:

- We provide a detailed analysis of the DStat ecosystem, including the features and functionalities of DStat services.
- We investigate the sources of the machines being attacked in DStat panels, showing the (abused) infrastructure on which these services rely.
- We analyze over 70,000 DDoS attacks targeting DStat systems, including multi-terabit-per-second events, providing a comprehensive overview of current DDoS threats.
- We explore the role of Telegram in facilitating the DStat ecosystem by collecting over 500,000 messages, showing the interactions between users and DStat service providers.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CCS '26, The Hague, Netherlands*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2871-6/2026/11
<https://doi.org/10.1145/3830454.3846638>

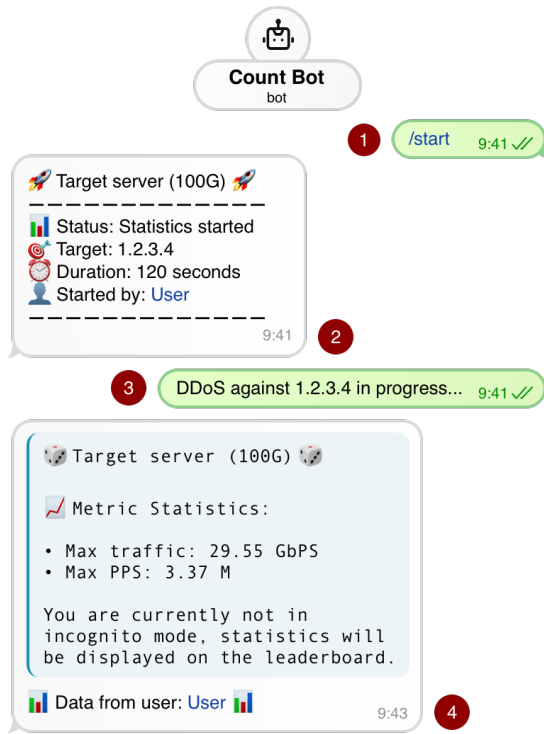


Figure 1: Steps taken to generate a Power Proof on Telegram.

2 Boasting DDoS capabilities

DDoS-for-Hire services offer a variety of attack methods, such as UDP floods, TCP SYN floods, and HTTP floods, among others [33]. Customers can choose the type of attack they want to launch, the duration of the attack, and the target IP address. The services typically operate through a web interface or a Telegram bot, allowing users to easily configure and initiate attacks. To attract customers, DDoS-for-Hire providers often boast their attack capabilities, such as the bandwidth they can generate or the number of packets per second they can send. From a customer perspective, these claims can be difficult to verify, leading to uncertainty about the potential impact of their chosen DDoS attack which they will only discover after the payment has been made.

Operators of DDoS-for-Hire services thus need a way to prove to their potential customers that they can deliver the attack power they promise. In this paper, we focus on the concept of *DDoS Power Proofs*, which are demonstrations of the attack power that a DDoS-for-Hire service can deliver. These proofs typically involve launching a DDoS attack towards a target system and measuring the resulting traffic to provide evidence of the service’s capabilities. These proofs are often shared on platforms like Telegram, where providers can reach a wide audience of potential customers.

2.1 Power Proof

The Power Proofs that we focus on in this paper are generated by interacting with a Telegram bot, which provides a specific endpoint to attack, and then posts the measured traffic against that endpoint in a publicly available Telegram channel. We show the four main

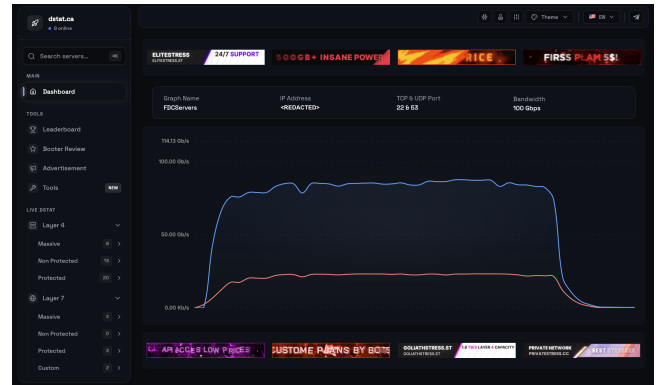


Figure 2: A screenshot of `dstat.ca`, as seen on March 27, 2026. An attack of ± 75 Gbps is visible.

steps of a user creating a DDoS Power Proof in Figure 1. First, the user starts a conversation with a DStat Telegram bot (step 1), after which it gives the user a duration and an endpoint to attack (step 2). The actor then starts a DDoS attack against the provided endpoint (step 3). Depending on the bot, a “live” feed is shown of the traffic received on the endpoint. When the attack time is finished, a full report is sent (step 4) and forwarded to a separate Telegram channel. This full report is called a Power Proof. These Power Proofs are shared on Telegram channels with many subscribers, and often include advertisements for the DDoS-for-Hire service that performed the attack. This allows providers to effectively market their services while demonstrating their capabilities to potential customers.

Generating Power Proofs on attacks is not solely used by DDoS-for-Hire providers, but also by individuals (or groups) who simply want to show off their attack power, without necessarily having a DDoS-for-Hire service to sell. This is especially common among users who have access to large botnets, and want to demonstrate their capabilities to the community. These individuals often share their Power Proofs on Telegram as well.

2.2 DStat platforms

Many DStat websites exist, functioning as platforms that aggregate data from a set of DStat endpoints and showing a live view of the incoming bandwidth. These DStat endpoints are DDoS attack targets which measure incoming traffic at the endpoint and report real-time statistics of this traffic. These endpoints differ in their specifications, with some having a maximum bandwidth of 1 Gbps, while others can handle over 1 Tbps. A screenshot of a DStat website can be seen in Figure 2, including a graph showing the incoming traffic to a DStat target over time.

Next to listing DStat targets and their network statistics, these websites offer a variety of other features. First, these websites serve as a place for DDoS-for-Hire providers to advertise their networks. The screenshot in Figure 2 shows banners on the top and bottom of the page, each being an advertisement for a booter. In addition, these websites often have a section for “booter reviews” or a leaderboard, where they list certain booters together with their ranking and potential power. Finally, almost all of these websites include a

Telegram link that goes to a group chat or Telegram bot related to the website. These channels are often used to share Power Proofs, and are tightly integrated with the DStat platform itself, allowing users to create Power Proofs straight from Telegram.

2.3 Telegram communities

The DDoS community extensively uses Telegram to share Power Proofs and talk about DDoS. Most DStat websites, as well as stresser providers, have their own dedicated Telegram groups that often have integrated bots to interact with the DStat platform or even with a stresser service to perform attacks right out of Telegram.

These bots generally have several features:

- Advertisements for Stresser providers
- Creating DDoS Power Proofs
- Daily or all-time leaderboards showing the strongest attacks (with links to individual Power Proofs)

As the bots share the generated Power Proofs in Telegram channels, these channels thus provide a “timeline”-like view of performed DDoS Power Proofs. Figure 1 shows an example interaction of a user starting and finishing a Power Proof within Telegram.

Next to these bots, the Telegram group chats are also used for “casual” chatting between users in the community, among which DDoS-for-Hire providers. These messages contain valuable information on how botnets are maintained, and help identify which users are important players in the ecosystem.

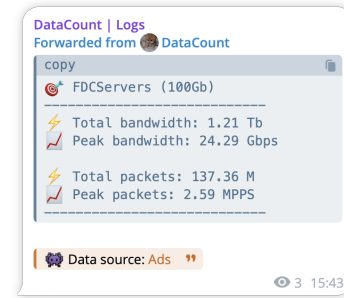
3 Scraping Telegram communities

As discussed in Section 2, the DStat community mainly uses Telegram as a social platform. That is why many of the available DStat websites contain a link to an associated Telegram chat. Within these chats, discussions arise about DDoS-for-Hire and DDoS Power Proofs. Additionally, these chats allow user to interact with the Telegram bots, discussed in Section 2.3.

We collect messages, user interactions, and shared links to DStat services. This allows us to gain insights into the popularity of different DStat services, user feedback, and the overall dynamics of the DStat ecosystem on Telegram. We also identify key influencers and operators within these groups who are affiliated with DStat or DDoS-for-Hire services.

To obtain a set of Telegram channels, we start by joining the groups linked on DStat websites. In these chats, we can find the Telegram bots used to generate Power Proofs. These bots often include advertisements with a link to a corresponding Telegram chat. Additionally, many users in these groups have Telegram links in their usernames or biographies to chats related to DDoS-for-Hire or DStat. This iterative process took place during the entire data collection period. Discovered and relevant Telegram chats were then added to the set of scraped chats.

In total, we monitor 41 distinct Telegram channels starting November 2025 end ending April 2026. Some channels include historical data, which is why our final Power Proof dataset starts in August of 2025. During our data collection period, we collect 505,065 messages sent by a total of 3,957 unique users.



(a) The Power Proof from the Telegram bot.

Property	Value
ChatID	3533428838
MessageID	6446
Timestamp	2026-04-06 13:43:56
User	Ads
EndpointName	FDCServers
EndpointCapacityRaw	100Gb
EndpointCapacityBps	100000000000
MaxTrafficBps	24290000000
MaxPacketRate	2590000

(b) The Power Proof parsed in a structured format.

Figure 3: An example of parsing a Power Proof message into a structured data format.

3.1 Parsing Power Proofs from Telegram messages

Every DStat Telegram bot generates a slightly different Power Proof, but they all share the same purpose: reporting the observed traffic of a DDoS attack against a DStat target. To obtain an overview of the performed DDoS attacks, and thus a general overview of DDoS capabilities, we parse these Power Proofs from the Telegram messages. We do this by creating regex rules for each of the bots that generate Power Proofs, which allow us to extract the relevant information from the messages and store it in a structured format for further analysis. The used regex rules are listed in Appendix A.2. In Figure 3, we show an example of a parsed Power Proof. In total, we were able to parse 70,390 DDoS Power Proofs with this methodology.

When Power Proofs are finalized, they do not always show the IP address of the DStat target, but only the names under which these targets are known. When users start their measurements from a private chat with a DStat bot (which we cannot observe), we cannot create a reliable mapping between Power Proofs and target IPs. To create this mapping, we rely on the first and last seen messages that map a target name (such as “FDCServers” in Figure 3) to an IP address based on public DStat bot interactions. Applying this methodology results in a subset of 23,504 attacks where we were able to map target IPs to Power Proofs.

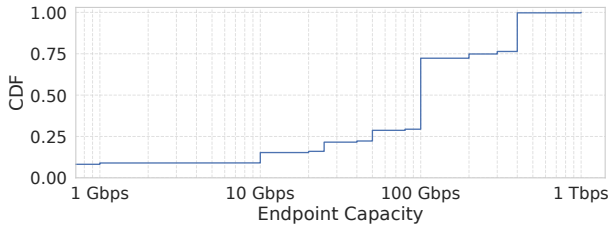


Figure 4: Distribution of endpoint capacities observed in DStat attacks. Targets with >1Tbps capacity are set to 1Tbps due to an unknown real limit.

4 The DStat ecosystem

Generating DDoS Power Proofs is straightforward, but it requires a certain infrastructure to be in place. As many of the endpoints used for Power Proofs are large in bandwidth, the cost of maintaining such infrastructure can be significant. There is however very little upside to hosting these DStat services, as they primarily serve to demonstrate attack capabilities rather than generate revenue. Advertisements on DStat websites cost around *40 US dollars* per month, based on conversations with DStat operators (see Appendix A.1).

4.1 Understanding DStat infrastructure

To understand the supporting infrastructure of DDoS Power Proofs, we investigate the endpoints that are used to monitor attacks, as they are at the core of what makes DDoS Power Proofs possible. By looking at the exposed services on these endpoints using Censys [11], we find that they all expose either Netdata [5] or Prometheus [6] openly to the Internet. These are services used for monitoring system resource usage, including the current network bandwidth usage. These services are largely unauthenticated (Netdata has a password mechanism, but browsing the metrics can be done anonymously), and are trivially scraped for data, making them perfect hosts to use for Power Proofs. Operators of DStat services can simply find these (legitimate) exposed monitoring endpoints, and use them as DStat targets for their Power Proofs. We can validate this by observing Telegram chats among DStat users and operators, which we talk about in more detail in Section 6.

While DStat websites often contain between 10 and 50 endpoints to attack, abusing legitimate servers as attack targets allows DStat services to pick from a large pool of potential targets, without having to invest in their own infrastructure. Searching for exposed Netdata and Prometheus instances on Censys, we find that there are around 50,000 and 1,000,000 such endpoints respectively. If endpoints notice that they are under constant attack and remove or firewall the monitoring tools, DStat services can simply find another exposed monitoring endpoint to use as a DStat target. We find that on average, DStat targets are listed on DStat websites for around 24 days¹, which means that DStat services need to find new targets every few weeks.

The capacity of listed DStat targets varies widely, with some having a maximum bandwidth of 1 Gbps, while others can handle up to

¹If we exclude DStat endpoints that were only observed once, we see an average of 35 days.

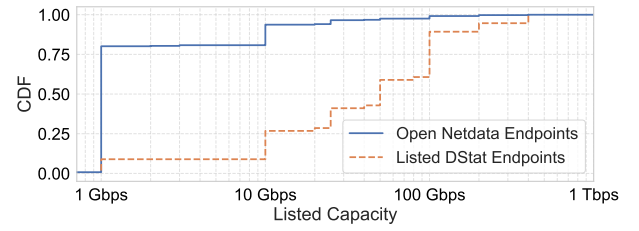


Figure 5: Capacity of listed DStat endpoints and open Netdata endpoints. Endpoints with large capacity (at least 10 Gbps) are way more likely to be used as DStat targets.

100 Gbps. Figure 4 shows the distribution of bandwidth capacities of DStat targets, based on the information provided on the DStat websites. During the course of our data collection, we observe a set of endpoints that have uplinks of more than 1 Tbps, which is an order of magnitude higher than the average DStat target. For some of the largest endpoints that we observe during our data collection, we find that instead of exposing Netdata or Prometheus, they are exposing SNMPv1 metrics. SNMP is a protocol used for network management and monitoring, and can also be used to monitor network bandwidth usage. The fact that these large endpoints are exposing SNMP metrics instead of Netdata or Prometheus suggests that they may be using different infrastructure such as border routers.

To understand the selection criteria for the public endpoints that are used as DStat targets, we look at the endpoints included in DStat websites and compare them to the full set of exposed Netdata endpoints. We specifically look at Netdata as one of the DStat websites shares a public list of 107,923 Netdata endpoints. Using Censys, we can confirm that these are indeed open Netdata instances. As we do not want to enumerate the bandwidth of the endpoints ourselves (this would involve bypassing the login screens of the Netdata instances, which we deem unethical), we rely on the bandwidth information provided on the DStat website listing the endpoints with their uplink capacity. Using the Power Proofs generated against these endpoints, we verify that the reported capacity appears to be correct. We visualize the distribution of bandwidth capacities of the DStat targets and the full set of Netdata endpoints in Figure 5. We find that endpoints are used as DStat targets if they have a high uplink capacity, which makes sense given that DDoS providers want to show off their largest attacks. In the set of DStat targets, 50% of the endpoints have a bandwidth capacity of at least 50 Gbps, while in the full set of Netdata endpoints, only 7% of the endpoints have a bandwidth capacity of at least 50 Gbps. This shows that DStat services are specifically selecting high-capacity endpoints to use as DStat targets for their Power Proofs, which allows them to demonstrate larger attacks and attract more customers to their DDoS-for-Hire services.

As a user can choose the DStat target when generating a Power Proof, we also investigate at the relationship between the capacity of DStat targets and the number of attacks performed against them. We find a moderate statistical relationship (Spearman correlation, $\rho = 0.37$, $p = 0.0007$) between the capacity and the total number

of performed attacks. This shows that, over time, higher-capacity endpoints tend to be attacked more often than lower-capacity endpoints. We attribute this to the fact that higher-capacity endpoints are more likely to be used for Power Proofs that show larger attacks, which are more likely to be shared on Telegram channels with many subscribers, which in turn leads to more attacks against these endpoints over time.

Provided that larger endpoints attract more attacks, we hypothesize that they are more likely to be patched as the owners of these endpoints are more likely to notice the attacks and take measures to prevent them. However, we find that the lifespan of DStat targets with a capacity of at least 100 Gbps averages around 35 days, which is not significantly different from the average lifespan of all DStat targets (slightly higher at 35.4 days). We also do not find any correlation between the capacity of DStat targets and their lifespan. We hypothesize that this is because even though larger endpoints attract more attacks overall, smaller endpoints are more likely to be saturated by the attacks, which makes them more likely to be noticed and patched.

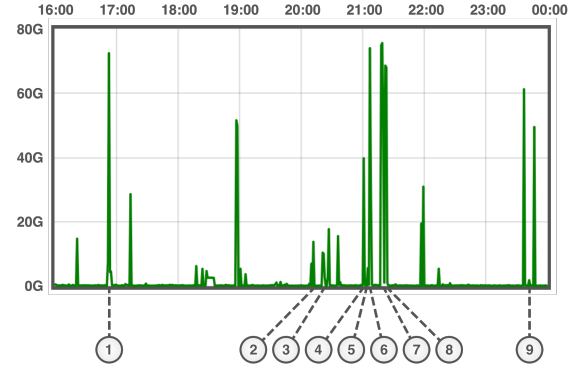
4.2 Attack validation

The published Power Proofs on Telegram are only useful for research if they correspond to real-world DDoS activity. While other members of the community would likely call out fake Power Proofs, it is still important to validate that the Power Proofs we observe on Telegram correspond to real attacks. First of all, we did not see any users mention anything about fake Power Proofs originating from DStat services in our Telegram dataset. However, we perform a small scale validation of the Power Proofs we observe on Telegram to ensure that they correspond to real-world DDoS activity.

To determine the validity of the Power Proofs, we compare these proofs against ground-truth data obtained from targeted systems. Since DStat endpoints abuse legitimate servers with exposed system monitoring tools, these same tools can be used to verify the integrity of the data shared on Telegram. However, to avoid causing additional load on already abused systems, we decided against scraping the targets automatically. Instead, we manually verify a subset of attacks across a selection of observed targets. We assume that, if we observe a 100% match between Telegram and the system logs on a substantial subset of the targets, the remainder of the attacks are also valid. Therefore, we sample 5% of all hosts observed as Power Proof targets, and manually verify several hours of data for each of these hosts, comparing the observed attacks in the system logs with the Power Proofs we observe on Telegram for the same time period. We find that for all of the sampled hosts, there is a 100% match between the attacks observed in the system logs and the Power Proofs we observe on Telegram, which gives us confidence that the Power Proofs we observe on Telegram correspond to real-world DDoS activity.

Figure 6 shows an example of how this manual validation is performed. In this example, shown in Figure 6a, we observe the ground truth data of a certain DStat endpoint from 16:00 until 23:59 on April 22nd, 2026. We use the following Prometheus query to get the incoming network traffic in bits per second:

```
irate(node_network_receive_bytes_total[1m]) * 8
```



(a) Prometheus metrics showing the measured bandwidth.

#	Timestamp	User	Bandwidth
1	2026-04-22 16:54	XRATsier [t.me/bootanet]	79 Gbps
2	2026-04-22 20:12	margielic traceguard	16 Gbps
3	2026-04-22 20:23	margielic traceguard	15 Gbps
4	2026-04-22 21:01	margielic traceguard	101 Gbps
5	2026-04-22 21:06	margielic traceguard	14 Gbps
6	2026-04-22 21:09	margielic traceguard	103 Gbps
7	2026-04-22 21:19	XRATsier [t.me/bootanet]	93 Gbps
8	2026-04-22 21:23	sidney	80 Gbps
-	2026-04-22 23:10	highraw	704 Mbps
9	2026-04-22 23:43	поддельщик	3 Gbps

(b) Observed Telegram Power Proofs for this target.

Figure 6: An example of manual attack validation for a DStat target. The time range of this example is from 16:00 to 23:59 on April 22nd, 2026. The <1Gbps attack is not matched for readability.

We compare this data to the Power Proofs we observe in our Telegram dataset for that endpoint in the same time range. These Power Proofs are shown in Figure 6b.

For this example, we see that all Power Proofs are represented in the ground truth data. The difference between the shown bandwidth in Prometheus and the reported peak bandwidth in the Power Proof can be explained by how the used query works. Specifically, our query applies the `irate` function to the received bytes counter over a 1-minute range. This estimates the per-second rate of incoming traffic based on the two most recent samples. Because this estimation depends on the sampling interval of the Prometheus node exporter, short-lived traffic peaks may be under- or overrepresented if they occur between scrapes.

Next to the matched attacks, we also see spikes in the ground truth data not represented in a Telegram Power Proof. This is likely due to the fact that the IP address of this target is also published on a DStat website. This means that anyone can perform a DDoS attack against this target and observe the resulting bandwidth, without using the Telegram bot and generating a Power Proof. This phenomenon emphasizes the lower-bound characteristic of our resulting dataset.

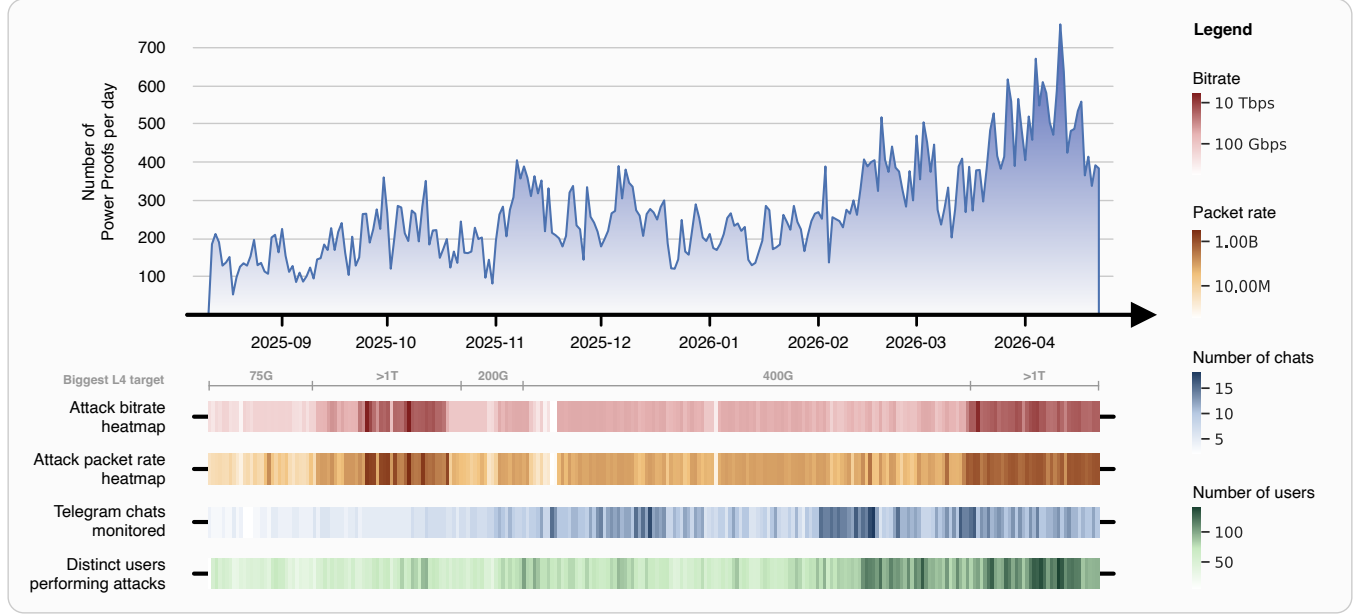


Figure 7: Overview of published Power Proofs.

5 Attacks

To understand the capabilities of DDoS operators that use Power Proofs to showcase their potential impact, we analyze the 70,390 Power Proofs shared in Telegram between August 2025 and April 2026. In this section, we report on the frequency and volume of the observed attacks, as well as the distribution of attack sizes.

5.1 Attack frequency

To understand the frequency of attacks on DStat endpoints, we analyze the number of Power Proofs observed per day during our monitoring period. Figure 7 shows a time series of the number of Power Proofs observed. On average, we observe 276 Power Proofs published per day, with an average of 75 unique users per day performing attacks. Users thus often publish multiple Power Proofs per day, potentially trying to get the best statistics possible. From the start of our monitoring period, we identify an increase in daily published Power Proofs, with a peak of 754 Power Proofs published on the 11th of April 2026. This increase in published Power Proofs could be due to a variety of factors, such as an increase in the number of DDoS-for-Hire providers using Power Proofs, an increase in the popularity of DDoS attacks, or an increase in the number of users interested in showcasing their DDoS capabilities. While we also increase the number of telegram chats we monitor during our monitoring period, the increase in published Power Proofs cannot be fully explained by this since we can historically scrape the channels used as a feed for performed Power Proofs.

As shown in Figure 7, the number of users performing attacks also increases over time, with a peak of 140 unique users on the 11th of April, 2026. Similarly to the increase in published Power Proofs, this increase in unique users cannot be fully explained by the increase in monitored Telegram chats. This suggests that there is indeed an increase in the number of users interested in

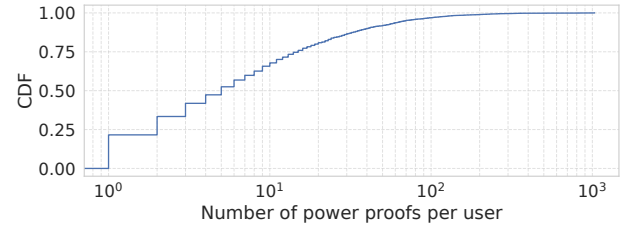


Figure 8: Distribution of Power Proofs per user.

showcasing their DDoS capabilities using Power Proofs. We show the distribution of the number of published Power Proofs per user in Figure 8. There is a small group of users that publish a large amount of Power Proofs, with the top user (“Mr.Kuru”) publishing 1,033 Power Proofs during our monitoring period. The distribution is long-tailed, with most users publishing only a few Power Proofs. The highest number of published Power Proofs in a day by a single user was 84 on the 11th of April 2026 by user “rolish”.

5.2 Attack volume

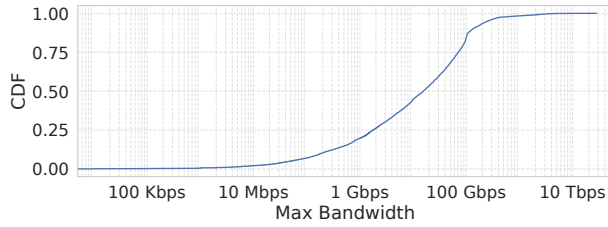
To understand the volume of the attacks performed for Power Proofs, we analyze both the peak bandwidth and peak packet rate. Figure 7 shows a timeline of attack bitrates and packet rates for all observed attacks during our monitoring period. To generate a large amount of traffic, attackers will have to send a large number of packets as well, so we observe a strong correlation between the two. However, there are some outliers that have a high packet rate but a relatively low bitrate, which are often directed at Layer 7 DStat endpoints, and have the goal to achieve high packet rates or bypass DDoS mitigation techniques.

Time	User	Bandwidth	PPS
2025-10-07 11:15	fox0000	27.98 Tbps	2.45B
2025-09-25 09:51	9gigsofram / @fox0000	25.20 Tbps	2.21B
2026-03-19 12:59	MSCTF32	18s.75 Tbps	0.71B
2026-03-19 10:32	MSCTF32	14.86 Tbps	0.95B
2026-03-19 12:52	MSCTF32	12.37 Tbps	0.90B
2026-04-05 14:51	Carl [satellitestress.st]	8.96 Tbps	0.91B
2026-03-19 12:16	Top Jew	8.01 Tbps	0.77B
2026-03-19 13:01	Goofyorg goofystress.st	8.01 Tbps	0.77B
2026-04-04 03:19	Carl [satellitestress.st]	7.94 Tbps	0.80B
2026-04-05 14:45	Carl [satellitestress.st]	7.91 Tbps	0.78B

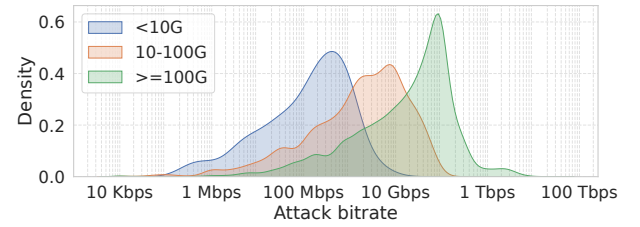
(a) Highest bitrate attacks.

Time	User	Bandwidth	PPS
2025-10-07 23:51	Kate Johnson	0.73 Tbps	3.14B
2025-10-07 19:39	x_x	0.72 Tbps	3.11B
2025-10-07 11:15	fox0000	27.98 Tbps	2.45B
2025-09-25 09:51	9gigsofram / @fox0000	25.20 Tbps	2.21B
2025-10-11 08:42	daytwo	0.48 Tbps	2.08B
2025-10-01 05:15	Thomas Kurian	0.41 Tbps	1.79B
2025-09-26 20:34	9gigsofram	7.73 Tbps	1.69B
2025-09-30 03:19	Thomas Kurian	0.37 Tbps	1.60B
2025-09-26 20:41	Thomas Kurian	0.36 Tbps	1.56B
2025-09-27 07:26	Thomas Kurian	0.36 Tbps	1.53B

(b) Highest packet-rate attacks.

Table 1: Top 10 attacks by peak bandwidth and peak packet rate.**Figure 9: CDF of observed attack bitrates.**

5.2.1 Attacks by bitrate. Table 1a shows the top 10 observed attacks based on their maximum bitrate. While there is a large spread in the observed attack bitrates, with a median of 10.66 Tbps and a mean of 13.99 Tbps, the top 10 attacks all have a bitrate exceeding 10 Tbps, with the highest observed attack having a bitrate of 27.98 Tbps spread over 2.45 billion packets per second. This means that the packets used in this attack have an average size of around 1440 bytes, which is a common packet size for DDoS attacks [12]. To measure these high bitrates, the attack needs to be directed at a DStat endpoint with a high uplink capacity, as otherwise it will saturate the endpoint's uplink. This means that the observed attack bitrates are influenced by the available DStat endpoints and their uplink capacities, and may not represent the true maximum power of the DDoS-for-Hire services performing these attacks. However, when there are DStat endpoints available with an uplink greater than 1 Tbps, we can start seeing true attack limits. Figure 7 notes the largest available uplink for a DStat endpoint at any given time, which shows that there are periods where the observed attack bitrates are not limited by the available DStat endpoints, and thus represent true attack limits. The distribution of all observed attacks is shown in Figure 9, showing that 50 percent of the attacks are smaller than 5 Gbps. There is a drop in the number of attacks after 400 Gbps, which corresponds to the 400 Gbps uplink that was observed for a large portion of time. Due to the small volume of these high-capacity attacks, it is difficult to determine whether this drop is due to the available DStat endpoints or due to true attack limits of DDoS networks.

**Figure 10: Observed attack bitrates grouped by target endpoint capacity.**

To better understand the measurements of the DStat endpoints and the observed attack rates, we identify for every attack the DStat endpoint it is directed at, and the maximum available uplink for that endpoint at the time of the attack. The goal of this is to understand whether attackers choose their DStat endpoint based on the attack they believe they can generate, or whether they just choose any available DStat endpoint and generate an attack that saturates it. The attack sizes measured by endpoints of different sizes are shown in Figure 10. As shown in the figure, many attacks get close to saturating the DStat endpoints they are directed at, and therefore there is only a lower bound on the true attack power of these attacks. However, when there are DStat endpoints available with an uplink greater than 1 Tbps, we can start seeing true attack limits, as there are many attacks that do not saturate these high-capacity endpoints. Looking at the attack distribution during the time a practically unlimited DStat endpoint was available, we are able to observe the most clear attack limits within the population. This distribution is similar to the overall distribution, and therefore we can conclude that the overall distribution is not heavily influenced by the available DStat endpoints, and thus represents a good estimate of the true attack power in these attacks.

5.2.2 Attacks by packet rate. Table 1b shows the top 10 observed attacks based on their maximum packet rate. Similar to the attack bitrates, there is a large spread in the observed attack packet rates. With a median of 1.93B packets per second and a mean of 2.12 billion packets per second, the top 10 attacks all have a packet rate greater than 1.5 billion packets per second. The highest observed attack

Country	Observations	Weighted traffic share
Brazil	1,776	31.64%
United States	1,934	20.89%
India	1,050	7.13%
France	1,144	3.73%
Argentina	1,126	3.38%
Colombia	1,141	3.32%
Indonesia	453	2.63%
Viet Nam	244	2.55%
Ukraine	1,021	2.39%
Mexico	169	2.33%

Table 2: Top 10 countries based on DDoS attack traffic sources.

has a packet rate of 3.14 billion packets per second and a bandwidth of 0.73 Tbps. This means that this attack has an average packet size of only 185 bytes. This suggests that the attack may be designed to achieve high packet rates or to bypass DDoS mitigation techniques that rely on detecting high volumes of traffic with larger packet sizes. We see however in Figure 7 that the packet rate correlates with the bitrate, as an attacker needs to send large packets when performing a high bitrate attack as well.

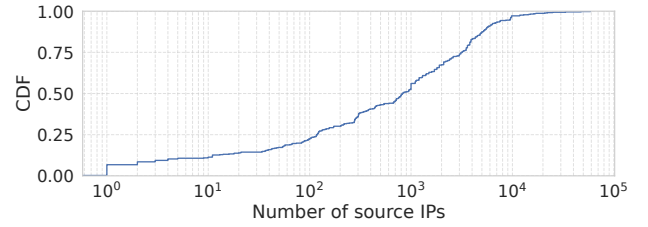
Specific endpoints exist for testing packet-rate attacks, some of which are located behind an anti-DDoS service (such as Cloudflare). These endpoints are often labeled “protected” and serve as a test to check whether an attack would circumvent DDoS mitigation. We expand on these cases in Section 5.4.

5.3 Attack characteristics

To further understand the nature of the attacks performed to generate Power Proofs, we analyze various characteristics of the attacks, including their duration, protocols used, and whether they employ IP spoofing.

When using a Telegram bot to generate a Power Proof, the measurement duration is typically specified by the bot, and users cannot choose any duration they want. This means that for many of these Power Proofs, the attack will last between one and two minutes. A user could attack for a longer time, but this is no longer saved and reported on in the Power Proof. Depending on the Telegram bot used, detailed reports including IP protocols, source ports, sources ASes and source countries are also generated. Figure 14 in Appendix A.4 shows an example.

In our dataset, we have 2,539 messages that show used attack types. From these, 2,444 (96.2%) attacks contain a UDP element, 536 (21.1%) contain a TCP element, and 403 (15.8%) contain an ICMP element. This is in line with previous studies on DDoS attacks [36], which have shown that UDP is a commonly used attack vector for volumetric attacks. Next to this, our dataset contains 2,520 Power Proofs that show source countries of the attack traffic. Table 2 shows an overview of the top 10 countries from which DDoS attack traffic originates. In this table, the number of observations indicates how many Power Proofs we see that show traffic originating from that country. The traffic share is weighted, with the weight of an attack based on the peak packet rate. This ensures that the data reflects the share of traffic seen from the country in question. We observe

**Figure 11: Number of unique source IPs observed in performed DStat attacks.**

the biggest share of traffic originating from Brazil, which is in line with previous studies and reports on DDoS botnets [1, 8, 35].

Finally, our dataset contains 457 Power Proofs that show the number of source IPs observed during a DDoS attack. These Power Proofs do not measure bandwidth, but consider only packet rate and number of source IPs. The used endpoint for this is named “IP Count”. Figure 11 shows a cumulative density function reflecting the number of source IPs seen in these Power Proofs. With an average of 2,382 sources and a median of 827, we observe that many attacks are performed using relatively small networks. We observe 13 cases where the number of source IPs exceeds 10,000, explaining the large difference between the median and average values.

To more accurately determine whether attacks employ IP spoofing, we rely on a separate dataset collected from a network telescope that monitors backscatter traffic as per the method of [25]. By correlating the timing and IP addresses of the attacks observed on the DStat endpoints with the backscatter data, we can identify whether an attack performed IP spoofing. Note that this will provide a strict lower bound on the number of spoofed attacks, as the spoofed attack traffic needs to hit the telescope’s monitored IP space to be observed. Next to this, the limitation described in Section 3.1 forces us to derive this statistic from a subset of our full attack dataset. We rely on a telescope with approximately 65,000 IP addresses, which covers $\approx 1/2^{16}$ of the entire IP space. In our data, we find 339 attacks showing evidence of IP spoofing, all of them using the TCP protocol. This, together with the observed number of source IPs from the “IP Count” DStat target, shows that the vast majority of volume-based attacks do not make use of IP spoofing. This corresponds to recent studies of DDoS botnets, where IP spoofing is not necessarily part of the attack logic [37].

5.4 Stress-testing DDoS mitigation

While most Power Proofs provide basic metrics for both bandwidth and packet rate, specific Power Proofs exist for layer 7 attacks against endpoints protected with DDoS mitigation. These Power Proofs usually show the total amount of requests, as well as the ratio of allowed and blocked packets. An example output of such a Power Proof looks as follows:

```
“Metric Statistics:
• Allowed - 19.51 M (56.08%)
• Blocked - 15.28 M (43.92%)”
```

These Power Proofs essentially allow users to check how effective a certain DDoS attack is against targets located behind a

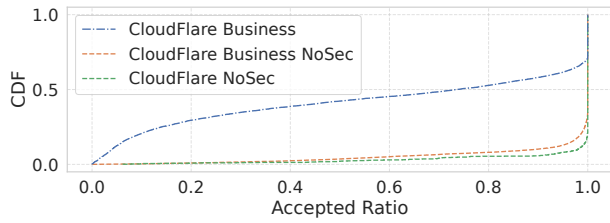


Figure 12: A CDF of accepted ratios observed in Cloudflare Power Proofs. NoSec indicates reduced security rules on the DStat target.

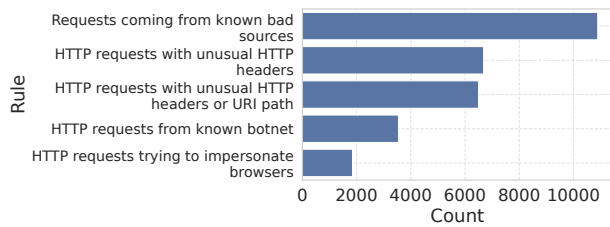


Figure 13: Top 5 most commonly matched Cloudflare mitigation rules in Telegram Power Proofs.

DDoS mitigation provider. Additionally, they allow botnet owners to check their effectiveness and update the attack vectors accordingly. In our data, we observe a total of 22,970 Power Proofs that show these metrics. All of these Power Proofs target Cloudflare specifically. Figure 12 shows a cumulative density function showing the accepted ratios observed in Cloudflare Power Proofs based on the listed protection levels. An accepted ratio of 0 means that all packets were blocked, while an accepted ratio of 1 means that all packets were allowed through.

When we only consider the Power Proofs where the target had Cloudflare DDoS mitigation enabled (Cloudflare Business), we see a total of 5,807 Power Proofs. Within these, we see 1,673 (28.81%) reports where no packets are dropped, which are mainly small attacks that do not trigger any DDoS mitigation rules. Through Telegram messages we can see that these attacks are often used for testing whether the attack is working at all, and to check the baseline performance of the attack. Another testing feature of these Power Proofs is the ability to see which mitigation rules are matching against the traffic. An example of a Power Proof, where this information is shown, can be seen below:

“Cloudflare Business NoSec (∞)

Metric Statistics:

- *Total requests: 140.57 M*
- *Allowed - 75.11 M (53.43%)*
- *Blocked (Requests coming from known bad sources, HTTP requests from known botnet (signature #22), HTTP requests with unusual HTTP headers or URI path (signature #56)) - 65.47 M (46.57%)”*

Chat name	Subs	Act	Act+	Msg
DstatST Chat	1,273	840	768	135,645
Käseküche	602	537	531	76,675
DstatST Chat	1,200	703	664	70,996
DstatCount	883	533	480	33,286
DstatRU	167	175	175	30,287
DataCount Chat	581	475	400	20,848
Dstat.SU Chat	378	461	456	19,425
DataCount Chat	130	72	57	3,209

Table 3: DStat-related chats sorted by number of messages. Subs = peak number of members, Act = active users, Act+ = active users engaging in conversation, Msg = total messages. Duplicated chat names indicate distinct Telegram chats.

In our data, we see a total of 12,464 Power Proofs where at least one Cloudflare DDoS mitigation rule is present. Figure 13 shows an overview of the top 5 most commonly observed Cloudflare mitigation rules in these Power Proofs. We observe that “requests coming from known bad sources” is by far the most matched rule.

These Power Proofs allow users to test the effectiveness of their attacks against DDoS mitigation, and to adjust their attack vectors accordingly before performing an attack against a real target. The readily available testing environment lowers the barrier for DDoS attackers to test and improve their attacks, making the ecosystem potentially more dangerous. We detail conversations about the testing in Section 6.2.4.

6 Telegram communities

As shown in Section 2, Telegram is widely used as a platform for communication and promotion of DDoS-for-Hire services. Next to this, many DStat platforms have dedicated publicly-available Telegram chats and channels. Here users can discuss their experiences, share DDoS Power Proofs, and promote their services. During our study, we monitor a total of 41 different chats that are related to DStat platforms or DDoS-for-Hire services. Across these chats, we see 505,065 messages sent by 3,957 unique users. This large number of messages underpins the extensive use of Telegram as a communications platform.

6.1 Community size

To get more insights into the community behind DStat specifically, we consider a subset of chats based on their activity and relevance to DStat services. More specifically, many of these chats are directly related to an available DStat website. Table 3 shows an overview of the considered chats. We distinguish between users that are active in the chat by sending at least one message (Act), and the subset of users that are active in conversation by commenting or replying to messages (Act+). Note that the number of active users can be higher than the number of subscribers, as users can leave the chat after sending a message. The number of messages (Msg) is the total number of messages sent in the chat during our monitoring period. We find that on average, 47% of members per chat actively engage in conversation, while 53% of all users never interact with other users and only read the messages in the chat. When we consider the top

Username	Message count	Affiliation
Juuvinile	8609	-
North	6526	valtrix.ru
mullbach	6417	returnstress.st
maxthecar	6276	-
DenilWD	5141	webd.st
JuggasBack	4703	bigfemboypower
Kazi	4590	-
telnetbruterthat...	4295	-
Censored	4277	dstat.st
tumult_1337	4253	skybyte.cloud

Table 4: Top 10 users in DStat-related chats based on message count.

Keyword list	Number of messages
Botnet infrastructure	7,449
Attack capabilities	17,789
Commercialisation	6,474

Table 5: Number of matched messages for the keyword lists provided in Appendix A.3.

10 users (excluding bots) in terms of message count, we find they are responsible for around 14% of all messages sent across these chats. Table 4 shows an overview of the top 10 users (non bots), including their affiliations in their username or user biography.

The dominance of DDoS providers among highly active users and the abundance of passive “lurking” users shows the dynamic of these Telegram chats: potential clients observe silently while service providers actively engage in conversations, share DDoS Power Proofs, and promote their services. This dynamic is further reinforced by the competitive nature of these chats, where users often compare their attack capabilities and seek validation from their peers. We investigate this competitive nature in more detail in the next section.

6.2 Exploring the dynamics of Telegram chats

To explore in what way users interact in these Telegram chats, we perform a keyword-based search of messages. We split the full set of messages into different categories, each with their own corresponding set of keywords used for search. The full list of keywords, generated with an LLM, can be found in Appendix A.3.

Table 5 shows the amount of messages we match for our three provided keyword lists. While it is challenging to ensure the context of these matches without considerably more complex algorithms, it gives us a general indication of what the community likes to discuss in the Telegram chats. We see that “attack capabilities” is the most matched category, which makes sense given that the main goal of the DStat community is to provide users with the option to experiment with and show attack power and capabilities.

To get an idea of how the community interacts in general, we consider five categories. First, we look for (1) information on attack infrastructure and (2) collaboration and asking for help using

our “botnet infrastructure” keyword list. Next to this, we look for (3) competition for who has the most power and (4) improving attack infrastructure with our “attack capability” keyword list. Finally, we look for (5) users looking to buy a DDoS attack with our “commercialisation” keyword list.

6.2.1 Attack infrastructure. In many Telegram chats, users talk about creating and maintaining their botnets, and also like to boast about their botnet size, showing the number of bots connected to their Command and Control server. For example, they talk about which exploits they use to build their networks and give advice to people starting their own network. Given that the overall goal is to build the largest and most powerful botnet, it is surprising to see that users are “helping” other users to build their networks:

The following conversation took place on November 16, 2025 in one of our monitored Telegram channels:

A: “What are you loading?”
 B: “Camera’s.”
 A: “Oh.”
 B: “23k cams, 1k routers and 2k fridges I think.”
 A: “What’s the output?”
 B: “2Tbps on the cams by themselves, and 3Tbps with all devices.”

The conversation continues with user A complaining about being limited by their budget. User B gives advice on which loading infrastructure they bought and how effective it was:

B: “Just abuse [Name of specific hoster].” “We loaded like 40k bots just with their servers.”

The information that is being shared in these chats can motivate new community members to start experimenting with their own networks. This can in turn increase the competition between DDoS-for-Hire providers even further.

Next to talking about how many bots people have in their networks, the location of the bots and the devices that comprise a botnet also matter to the community:

A: “Who got the most bots in here?”
 B: “No one.”
 A: “Looks like @*** has most bots and then after that its @***.”
 “But they are all loading android tvs :(.” “Brazil power s*cks.”
 “Show me some USA power.”
 C: “Go load TVT.” “You will find a lot of USA power there.”
 D: “Raysharp.” “I used to have 10k Raysharp.”
 A: “Proof.”

This ecosystem of sharing information on which devices to target and where to host loading infrastructure can lead to a more efficient DDoS-for-Hire market, as providers can learn from each other and optimize their services based on shared knowledge. This raises concerns about the ease with which new actors can enter the market and potentially contribute to an increase in DDoS attacks globally.

6.2.2 Collaboration and asking for help. We also observe instances of users asking for help and collaborating with each other. Specifically new users reach out to the community for help with setting up their botnet, and more experienced users give advice on how to do this. For example, a user who is new to using Mirai for building a botnet asks for advice on how to set up their network:

A: “I use mirai, I’m new you know.”
 B: “Don’t use telnet.”

C: “Low bandwidth, and tons of honeypots.” “I dont suggest using ur main cnc ip for bots to connect to.”
 A: “I use a proxy.”
 C: “Yeah that’s good.”

Here, users are not only sharing which protocols to target to grow their botnet, but also how to avoid detection by not triggering honeypots and using proxies to hide the Command and Control server. This type of collaboration can help new actors to enter the market and stay under the radar.

6.2.3 Competitive about power. Even though users regularly share techniques for building networks, the atmosphere in the chats remains dominantly competitive. Users often compare the size of networks, as well as the attack power. Within this competition, users often ask for a proof of power when other users make claims.

A: “I got a ton of dvr” “All 4 core cpus”
 B: “Show brother” “I wanna see proof” “All talk no proof”
 [A then performs a Power Proof showing an attack of 30 Gbps]

6.2.4 Improving attack infrastructure. Besides giving real-world measurements of how big DDoS attack are, Power Proofs can also be used by actors to improve their attack strategies. Power Proofs as described in Section 5.4 can be especially helpful to determine what needs to change to make an attack more effective. We can observe cases of this happening in our dataset. For example, a Telegram user used these Power Proofs to find the optimal rate-limit for their DDoS botnet. This conversation took place on April 8, 2026:

A: “I’m seeing how different rate limit effects bot.” “Maybe we break 300M packets on the next one.”
 [A then performs a Power Proof showing 259M allowed requests and 36M blocked requests matching Cloudflare signature #56]
 A: “Oh no they’re flagging at this much traffic.” “Ok we’re breaking 200.”
 B: “So many signature #56.” “Get better speed.”
 A: “Will be worked out by tomorrow.”

The information gained from the Power Proof allows this user to optimize their attack strategy, and by sharing this information in the Telegram chat, they can also influence other users to optimize their attack strategies as well.

6.2.5 Users looking to buy. Another use-case for these Telegram chats is users looking to buy DDoS attack capabilities from a DDoS-for-Hire provider. When looking for a DDoS-for-Hire provider, users often seek a test attack, likely to test the power of a DDoS-for-Hire provider in the real world.

A: “Who got big power and have test plans?”
 B: “For what?”
 A: “Layer 4”
 C: “DM me”

Additionally, DDoS-for-Hire provides often advertise the possibility for a test attack:

“Dm @presidentofnetherland for test hits.”

Messages like this get sent multiple times per day. For this reason, many groups have a rule that allows only one of these messages per user per day, including a warning in their welcome message:

“5. Only 1 (one) ‘DM for test hit’ message per day. Sending more will result in a daily mute”

6.3 Limitations of the Telegram dataset

In total, we collect over 500,000 messages from Telegram chats related to DDoS and Power Proofs. While this dataset has the potential to reveal interesting social dynamics of the DDoS-for-Hire community, the unstructured nature of chat messages and the informal language used makes a detailed analysis out of scope.

7 Related work

Understanding the DDoS landscape has been a focus of research for many years, with various studies analyzing attack patterns, motivations, and impacts [19, 26–28]. In this paper, we provide a novel perspective on the capabilities of the DDoS ecosystem by measuring ‘testing’ attacks.

Measuring DDoS attacks has traditionally relied on data from network telescopes [9, 14, 19], honeypots [15, 23, 29], and ISP logs [21]. These methods provide valuable insights but often suffer from limitations such as incomplete coverage and potential biases in the observed data, as shown by Hiesgen et al. [16]. Measuring the DDoS ecosystem is hard as there are many different attack types, which are geographically distributed, and often short-lived.

Kramer et al. [23], Griffioen et al. [15], and Rossow [29] focus on specific types of DDoS attacks, such as amplification attacks, which are easier to measure due to their distinct characteristics. Griffioen et al. [14] analyze the DDoS attack landscape by examining the distribution of attack sizes and durations of TCP SYN floods. Nawrocki et al. [27] investigate DDoS attacks by looking at DNS amplification attacks. Jonker et al. [19] aims to approach this problem by analyzing a large dataset of DDoS attacks collected from various sources, but even with this large dataset, the capabilities of the DDoS ecosystem are hard to measure.

The DDoS-for-Hire ecosystem has been studied from various angles. Hutchings et al. [17] explore the provisioning of DDoS-for-Hire services. Santanna et al. [30] provide an analysis of the websites used to offer DDoS-for-Hire services. Karami et al. [20] investigates the business models of DDoS-for-Hire services and the effectiveness of taking down these services. Kopp et al. [22] also analyze the effectiveness of taking down these services, and find that there is no significant impact on attacks when a website is taken down. Brunt et al. [10] investigate the effect of payment intervention against DDoS-for-Hire services, but recently the use of cryptocurrencies has made this type of intervention less effective. Vu et al. [36] show that taking down DDoS-for-Hire services can make an impact on the attack landscape, but that the effect is short-lived as these services quickly reappear. Hyslip and Holt [18] analyze the capacity of these DDoS-for-Hire services in cybercrime markets.

A major focus of previous works is the underlying infrastructure used in DDoS attacks, often botnets such as Mirai [8] and its variants [13]. Affinito et al. [7] analyze the evolution of Mirai and its variants over the last years, while Weyns et al. [37] show the current state of a Mirai variant, continuing to cause DDoS attacks.

This paper expands the understanding of the DDoS ecosystem as a whole by analyzing the attack capabilities of the ecosystem, rather than focusing on specific attack types or infrastructure. By measuring ‘testing’ attacks, we provide insights into the capabilities of the DDoS ecosystem that are not captured by traditional measurement methods.

8 Discussion

The DDoS ecosystem is complex, and has been around for a long time. Next to the sheer infrastructure and attack capabilities of DDoS providers, the ecosystem also has a social aspect, with Telegram groups being an important part of the community. The addition of a ‘trusted’ environment for DDoS providers to showcase their capabilities has significant implications for the ecosystem, as it allows providers to demonstrate their power in a way that is more credible than just sharing screenshots of websites that are down.

The effect of Power Proofs on the DDoS ecosystem. While we cannot measure the direct effect of Power Proofs on the DDoS ecosystem, we can speculate on the potential implications. The presence of a ‘trusted’ environment for DDoS providers to showcase their capabilities may lead to increased competition among DDoS providers, as they can now directly compare their attack power with others. This could potentially lead to an arms race, where providers continuously try to outdo each other in terms of attack power. While we do see evidence of this in Telegram channels, where users are boasting that their attacks are larger than those of someone else, we can not quantitatively show that there is an arms race based on our data. Additionally, the ability to demonstrate attack capabilities may also lead to increased demand for DDoS-for-Hire services, as potential customers can now see the actual power of the services they are considering, and may be more likely to purchase a service that has a strong track record of powerful attacks. We do see that in advertising efforts, DDoS providers often rely on Power Proofs to demonstrate their capabilities, which suggests that they see value in showcasing their attack power.

The underlying infrastructure. Exposed system monitoring tools are a key part of the DStat and Power Proof ecosystem, as they provide the infrastructure for generating Power Proofs. The fact that these tools are often exposed to the Internet without proper security measures in place poses a challenge, as these services are apparently not aware of the abuse, and do not take measures to prevent it. This calls into question whether we can classify these DStat targets as ‘victims’ of DDoS attacks, as they are not the intended targets of the attacks, and are instead being used as a means to an end for DDoS providers to showcase their capabilities. However, it is also important to note that these services are being abused without their consent, and may be experiencing negative consequences as a result of this abuse, such as increased bandwidth costs or degraded performance. We have coordinated with our national CERT to notify the owners of these abused services, and we hope that they will take measures to secure their infrastructure. However, the amount of open system monitoring endpoints is large and they will all need to be closed before DStat cannot abuse this technique anymore.

The DDoS ecosystem is open to help each other. The DDoS ecosystem is not just about competition, but also about collaboration and mutual support. We see evidence of this in Telegram channels, where users often share tips and advice on how to improve their attacks, or how to find new targets for Power Proofs. This suggests that while there may be competition among DDoS providers, there is also a sense of community and a willingness to help each other out. Disrupting these networks may therefore have good effects on the

ecosystem, as it may reduce the ability of providers to collaborate and share information, which could potentially lead to a decrease in the overall capabilities of the ecosystem.

Practical use of the resulting Power Proof dataset. While we think that Power Proofs are more harmful than beneficial, and we should aim to disrupt these networks, the presence of Power Proofs may also have implications for defenders and researchers. For defenders, the ability to see the attack power of DDoS providers may help them better understand the threat landscape and prioritize their defenses accordingly. For researchers, the availability of Power Proofs may provide valuable data for studying DDoS attacks and understanding the capabilities of different providers. For law enforcement, Power Proofs may offer insights into the capabilities and activities of DDoS providers, allowing them to better target their investigations to the networks that are most active or capable. However, it is important to note that Power Proofs may not always be representative of real-world attacks, as providers may choose to showcase their best attacks rather than a random sample of their attacks.

Future work. While this paper provides a comprehensive analysis of DStat services and Power Proofs, there are still many open questions and avenues for future research. One potential direction for future work is to investigate the economic aspects of DDoS-for-Hire services in more detail, such as the pricing models used by providers, how users are attracted to these services, and the overall market dynamics. Another area for future research is to explore the social dynamics of the DDoS community on Telegram, such as the role of influencers, the formation of sub-communities, and the impact of social interactions on the behavior of providers and customers. Additionally, the disruption of DStat services and Power Proofs could be an interesting area for future research, as it could provide insights into the resilience of the DDoS ecosystem and the potential impact of disrupting these networks.

9 Conclusion

In this paper, we present a comprehensive analysis of DStat services and Power Proofs, which provide real-time monitoring of DDoS attacks launched by DDoS operators. These services offer the unique ability for DDoS operators to showcase their capabilities in a ‘trusted’ environment. The DDoS community built a large ecosystem around these DStat services, which we investigate in detail. We find that DStat services rely on publicly exposed monitoring endpoints, allowing them to operate without significant investment in infrastructure. The owners of this abused infrastructure often seem unaware of the abuse, and do not take measures to prevent it. We investigate the attacks performed as Power Proofs, uncovering a wide range of attack sizes and characteristics that shed light on the capabilities of actors within the DDoS ecosystem. Furthermore, our analysis of Telegram groups shows the social dynamics and promotional strategies employed by DDoS-for-Hire providers. As a whole, the presence of a ‘trusted’ environment for DDoS operators to showcase their capabilities has significant implications for the DDoS ecosystem, and our findings provide valuable insights into the infrastructure, attack characteristics, and market dynamics of DDoS Power Proofs.

Acknowledgments

This work was supported by the Dutch Research Council (NWO) under the ADAPTive project and by the European Union under the Horizon Europe Programme as part of the project SafeHorizon (Grant Agreement #101168562).

References

- [1] 2025. Long Live The Vo1d Botnet: New Variant Hits 1.6 Million TV Globally. <https://blog.xlab.qianxin.com/long-live-the-vo1d-botnet/>
- [2] 2026. 2025 Q4 DDoS threat report: A record-setting 31.4 Tbps attack caps a year of massive DDoS assaults. <https://blog.cloudflare.com/ddos-threat-report-2025-q4/>
- [3] 2026. dstat(1) - Linux man page. <https://linux.die.net/man/1/dstat>
- [4] 2026. Global DDoS Threat Intelligence Reports | NETSCOUT. <https://www.netscout.com/threatreport/global-highlights/>
- [5] 2026. Netdata: Monitoring and troubleshooting transformed. <https://www.netdata.cloud/>
- [6] 2026. Prometheus - Monitoring system & time series database. <https://prometheus.io/>
- [7] Antonia Affinito, Stefania Zinno, Giovanni Stanco, Alessio Botta, and Giorgio Ventre. 2023. The evolution of Mirai botnet scans over a six-year period. *Journal of Information Security and Applications* 79 (2023), 103629.
- [8] Manos Antonakakis, Tim April, Michael Bailey, Matt Bernhard, Elie Bursztein, Jaime Cochran, Zakir Durumeric, J Alex Halderman, Luca Invernizzi, Michalis Kalitsis, et al. 2017. Understanding the mirai botnet. In *26th USENIX security symposium (USENIX Security 17)*. 1093–1110.
- [9] Norbert Blenn, Vincent Ghi  tte, and Christian Doerr. 2017. Quantifying the spectrum of denial-of-service attacks through internet backscatter. In *Proceedings of the 12th International Conference on Availability, Reliability and Security*. 1–10.
- [10] Ryan Brunt, Prakhar Pandey, and Damon McCoy. 2017. Booted: An analysis of a payment intervention on a ddos-for-hire service. In *Workshop on the Economics of Information Security*. 06–26.
- [11] Zakir Durumeric, David Adrian, Ariana Mirian, Michael Bailey, and J Alex Halderman. 2015. A search engine backed by Internet-wide scanning. In *Proceedings of the 22nd ACM SIGSAC conference on computer and communications security*. 542–553.
- [12] Eric Gamess and Brody Smith. 2020. Evaluation of TCP over IPv4 and IPv6 for the ESP8266 in Normal Operation and Under a DoS Attack. In *Proceedings of the 2020 ACM Southeast Conference*. ACM, Tampa FL USA, 243–247. <https://doi.org/10.1145/3374135.3385311>
- [13] Harm Griffioen and Christian Doerr. 2020. Examining Mirai’s battle over the Internet of Things. In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*. 743–756.
- [14] Harm Griffioen and Christian Doerr. 2020. Quantifying TCP SYN DDoS resilience: A longitudinal study of Internet services. In *2020 IFIP Networking Conference (Networking)*. IEEE, 217–225.
- [15] Harm Griffioen, Kris Oosthoek, Paul Van Der Knaap, and Christian Doerr. 2021. Scan, test, execute: Adversarial tactics in amplification DDoS attacks. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*. 940–954.
- [16] Raphael Hiesgen, Marcin Nawrocki, Marinho Barcellos, Daniel Kopp, Oliver Hohlfeld, Echo Chan, Roland Dobbins, Christian Doerr, Christian Rossow, Daniel R Thomas, et al. 2024. The age of DDoScovery: an empirical comparison of industry and academic DDoS assessments. In *Proceedings of the 2024 ACM on Internet Measurement Conference*. 259–279.
- [17] Alice Hutchings and Richard Clayton. 2016. Exploring the provision of online booter services. *Deviant Behavior* 37, 10 (2016), 1163–1178.
- [18] Thomas S Hyslip and Thomas J Holt. 2019. Assessing the capacity of DRDoS-for-hire services in cybercrime markets. *Deviant Behavior* 40, 12 (2019), 1609–1625.
- [19] Mattijs Jonker, Alistair King, Johannes Krupp, Christian Rossow, Anna Sperotto, and Alberto Dainotti. 2017. Millions of targets under attack: a macroscopic characterization of the DoS ecosystem. In *Proceedings of the 2017 Internet Measurement Conference*. 100–113.
- [20] Mohammad Karami, Youngsam Park, and Damon McCoy. 2016. Stress testing the booters: Understanding and undermining the business of DDoS services. In *Proceedings of the 25th International Conference on World Wide Web*. 1033–1043.
- [21] Daniel Kopp, Christoph Dietzel, and Oliver Hohlfeld. 2021. DDoS never dies? An XPP perspective on DDoS amplification attacks. In *International Conference on Passive and Active Network Measurement*. Springer, 284–301.
- [22] Daniel Kopp, Matthias Wichtlhuber, Ingmar Poesse, Jair Santanna, Oliver Hohlfeld, and Christoph Dietzel. 2019. DDoS hide & seek: on the effectiveness of a booter services takedown. In *Proceedings of the Internet Measurement Conference*. 65–72.
- [23] Lukas Kr  mer, Johannes Krupp, Daisuke Makita, Tomomi Nishizoe, Takashi Koide, Katsunari Yoshioka, and Christian Rossow. 2015. Amppt: Monitoring and defending against amplification ddos attacks. In *International Symposium on Recent Advances in Intrusion Detection*. Springer, 615–636.
- [24] Jelena Mirkovic and Peter Reiher. 2004. A taxonomy of DDoS attack and DDoS defense mechanisms. *SIGCOMM Comput. Commun. Rev.* 34, 2 (April 2004), 39–53. <https://doi.org/10.1145/997150.997156>
- [25] D Moore, C Shannon, D Brown, G Voelker, and S Savage. 2006. Inferring Internet Denial-of-Service Activity. *ACM Transactions on Computer Systems* 24, 2 (May 2006), 115–139. <https://doi.org/10.1145/1132026.1132027>
- [26] David Moore, Colleen Shannon, Douglas J Brown, Geoffrey M Voelker, and Stefan Savage. 2006. Inferring internet denial-of-service activity. *ACM Transactions on Computer Systems (TOCS)* 24, 2 (2006), 115–139.
- [27] Marcin Nawrocki, Mattijs Jonker, Thomas C Schmidt, and Matthias W  hlisch. 2021. The far side of DNS amplification: tracing the DDoS attack ecosystem from the internet core. In *Proceedings of the 21st ACM Internet Measurement Conference*. 419–434.
- [28] Arman Noroozian, Maciej Korczy  ski, Carlos Hernandez Ga  n, Daisuke Makita, Katsunari Yoshioka, and Michel Van Eeten. 2016. Who gets the boot? analyzing victimization by ddos-as-a-service. In *International Symposium on Research in Attacks, Intrusions, and Defenses*. Springer, 368–389.
- [29] Christian Rossow et al. 2014. Amplification hell: Revisiting network protocols for ddos abuse.. In *NDSS*. 1–15.
- [30] Jos   Jair Santanna, Ricardo O De Schmidt, Daphne Tuncer, Joey De Vries, Lisandro Z Granville, and Aiko Pras. 2016. Booter blacklist: Unveiling DDoS-for-hire websites. In *2016 12th International Conference on Network and Service Management (CNSM)*. IEEE, 144–152.
- [31] Jos   Jair Santanna, Romain Durban, Anna Sperotto, and Aiko Pras. 2015. Inside booters: An analysis on operational databases. In *2015 IFIP/IEEE International Symposium on Integrated Network Management (IM)*. 432–440. <https://doi.org/10.1109/INM.2015.7140320> ISSN: 1573-0077.
- [32] Jos   Jair Santanna and Anna Sperotto. 2014. Characterizing and Mitigating the DDoS-as-a-Service Phenomenon. In *Monitoring and Securing Virtualized Networks and Services*, Anna Sperotto, Guillaume Doyen, Steven Latr  , Marinos Charalambides, and Burkhard Stiller (Eds.). Springer, Berlin, Heidelberg, 74–78. https://doi.org/10.1007/978-3-662-43862-6_10
- [33] Jos   Jair Santanna, Roland van Rijswijk-Deij, Rick Hofstede, Anna Sperotto, Mark Wierbosch, Lisandro Zambenedetti Granville, and Aiko Pras. 2015. Booters — An analysis of DDoS-as-a-service attacks. In *2015 IFIP/IEEE International Symposium on Integrated Network Management (IM)*. 243–251. <https://doi.org/10.1109/INM.2015.7140298> ISSN: 1573-0077.
- [34] Gaurav Somani, Manoj Singh Gaur, Dheeraj Sanghi, Mauro Conti, and Rajkumar Buyya. 2017. DDoS attacks in cloud computing: Issues, taxonomy, and future directions. *Computer Communications* 107 (July 2017), 30–48. <https://doi.org/10.1016/j.comcom.2017.03.010>
- [35] Rui Tanabe, Tatsuya Tamai, Akira Fujita, Ryoichi Isawa, Katsunari Yoshioka, Tsutomu Matsumoto, Carlos Ga  n, and Michel van Eeten. 2020. Disposable botnets: examining the anatomy of IoT botnet infrastructure. In *Proceedings of the 15th International Conference on Availability, Reliability and Security (ARES ’20)*. Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3407023.3409177>
- [36] Anh V Vu, Ben Collier, Daniel R Thomas, John Kristoff, Richard Clayton, and Alice Hutchings. 2025. Assessing the Aftermath: the Effects of a Global Takedown against {DDoS-for-hire} Services. In *34th USENIX Security Symposium (USENIX Security 25)*. 3595–3612.
- [37] Maarten Weyns, Dario Ferrero, Stefan Op de Beek, Daniel Wagner, Georgios Smaragdakis, and Harm Griffioen. 2026. From Mirai to Gorilla: Deep Dive into a Long-Lasting DDoS-for-Hire Botnet. In *USENIX Security Symposium*. Baltimore, MD.

Ethical Considerations

In this work, we investigate the DDoS ecosystem by analyzing the infrastructure behind DStat services, the attacks performed on their endpoints, and the market dynamics of DDoS Power Proofs on Telegram. As this research involves DDoS attacks, which are harmful and illegal activities, we have taken several ethical considerations into account to ensure that our research is conducted responsibly and does not contribute to the harm caused by DDoS attacks. In this section, we discuss our considerations and the impact of our research on various stakeholders.

Telegram users: As our research involves analyzing Telegram chats, we collect data from public Telegram channels. These channels might contain DDoS providers, potential clients, researchers or law enforcement, and other users who are interested in DDoS attacks. To maintain the privacy of these users, we do not share

any identifiable information about them, such as their usernames. For the DDoS providers however, we do share their username in the paper if they (1) belong to the top “worst offenders” in terms of messages, DDoS power, or number of attacks, or (2) are directly involved in the attacks against DStat infrastructure discussed in the paper. As these users perform DDoS attacks publicly in open channels, we consider this information to be public and relevant for the research community. We ensure that we do not share any information that could lead to the identification of potential clients or other users who are not directly involved in DDoS attacks.

Open Netdata and Prometheus endpoints: Our research involves analyzing the infrastructure behind DStat services, which includes publicly exposed monitoring endpoints. We only analyze data from these endpoints that is publicly shared in DDoS Power Proofs or DStat panels, and do not access the endpoints directly. The only exception is that we access 5 endpoints of DStat services to verify that they are indeed publicly exposed and to understand how they work. We make sure to not add any additional load to these endpoints and only query them once.

We notified our national CERT about these endpoints, and they have disclosed this information to a set of relevant parties, and are working on disclosing this information further. We aim to spread awareness about the existence of these endpoints and the risks they pose, and we hope that this research can contribute to the mitigation of DDoS attacks by raising awareness about the capabilities of the DDoS ecosystem.

DDoS Mitigation Providers: Our research provides insights into the capabilities of the DDoS ecosystem, which can be valuable for DDoS mitigation providers. By understanding the attack patterns and capabilities of DDoS providers, mitigation providers can develop more effective strategies to defend against these attacks. We show that DDoS providers often test their attacks against DDoS mitigation providers, and we make sure to only share information about these tests that is already publicly shared in DDoS Power Proofs or DStat panels.

Decision to publish

We believe that the benefits of publishing this research outweigh the potential risks. By shedding light on the complexity of the DDoS ecosystem and the capabilities of DDoS providers, we can raise awareness about the risks of DDoS attacks and contribute to the development of more effective mitigation strategies. By identifying the infrastructure behind DStat services and disclosing this through publication, we hope that the Netdata and Prometheus communities can take action to secure their software and prevent abuse by DDoS providers. We also hope that this research can contribute to the broader understanding of the DDoS ecosystem and the dynamics of DDoS-for-Hire services, which can inform future research and policy decisions in this area.

Open Science

To show our support for the Open Science principles and make our research as transparent as possible, we share the main artifacts used in this work.

Specifically, we share (1) the full dataset of 70,390 parsed Power Proofs from Telegram and (2) the set of observed target IP addresses

mapped to DStat target names, together with the first and last seen timestamps. The majority of the results presented in this paper are derived from these datasets. These published datasets are publicly available and described in more detail at <https://zenodo.org/records/19705504>.

We decided against sharing the full dataset containing over 500,000 Telegram messages, since we cannot provide any guarantee that these messages do not contain any personally identifiable information. Nevertheless, we are open to share this data with trusted third parties upon request.

A Appendix

A.1 Advertising on DStat websites

The following Telegram conversation took place on September 1st, 2025. We asked the owner of `dstat.one` how expensive an advertisement spot is.

Us: “Hi! I am interested to buy an ad spot on `dstat.one`. What is the price?”

Them: “40\$ banner” “50\$ crown” “25\$ one time for listing” “50\$ lets count bot”

Us: “Thanks!”

Them: “all prices except of the listing is monthly ^^”

A similar conversation took place on September 2nd, 2025. This time with the owner of `dstat.love`.

Us: “Hey! I would like to buy an ad spot on `dstat.love`. What is the price?”

Them: “hi” “40\$ for banner”

Us: “Is that per month?”

Them: “ye”

A.2 Regex rules to parse Power Proofs

Every DStat Telegram bot generates a slightly different Power Proof. To accomodate this, we created several regex rules to parse as many messages as possible in our dataset.

For messages posted in chat with ID 2422133902:

```
Header:
"#\d+\s+(?P<endpoint_name>.*?)\s*\((?P<capacity>[^\)]+)\)"

Max bitrate:
"Max traffic:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[TGMK])\s*(?:\s*bps)?\b"

Max packet rate:
"Max PPS:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])\s*(?:\s*pps)?\b"

Total requests:
"Total requests:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])\s*(?:\s*bps)?\b"

Allowed requests:
"Allowed\s*\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])\s*(?P<ratio>\d+(?:\.\d+)?)%\s*"

User:
"Data from user:\s*@(?:P<user>[^\r\n]+)"
```

For messages posted by user with ID 8558868742:

```
Header
"#\d+\s+(?P<endpoint_name>[^()]+\s*\((?P<capacity>[^)]+\s*)\)"

Max bitrate:
"Max traffic:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[TGMK])?(?:\s*bps)?\b"

Max packet rate:
"Max PPS:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*pps)?\b"

Total requests:
"Total requests:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*bps)?\b"

Allowed requests:
"Allowed\s*-\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*pps)?\b"

User:
"Data from user:\s*@(P<user>[^\r\n]+)"
```

And finally, for messages posted by user with ID 8522392432:

```
Header:
"[emoji]\s*(?P<endpoint_name>.[?]+\s*\((?P<capacity>[^)]+\s*)\)"

Max bitrate:
"Peak bandwidth:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[TGMK])?(?:\s*bps)?\b"

Max packet rate:
"Peak packets:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*pps)?\b"

Total requests:
"Total packets:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*bps)?\b"

Allowed requests:
"Allowed:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[BMK])?(?:\s*pps)?\b"

Total bandwidth:
"Total bandwidth:\s*(?P<value>\d+(?:\.\d+)?)\s*(?P<unit>[TGMK])?(?:\s*b)?\b"

User:
"Data source:\s*(?P<source>[^\r\n]+)"
```

A.3 Keyword search in Telegram

We used four different keyword lists to find relevant information in our Telegram dataset. All the matching is done case-insensitive.

The first list concerns botnet infrastructure. This matching finds messages that contain any information on botnets, infections, exploits, etc.

```
['botnet', 'bots', 'zombies', 'nodes', 'infected',
'compromised', 'payload', 'c2', 'c&c', 'command and control', 'cnc', 'loader', 'dropper', 'spreading',
'propagation', 'exploit']
```

The second list concerns attack capabilities. This matching finds any messages that talk about attack power, different types of attacks, etc.

```
['bps', 'gbps', 'tbps', 'pps', 'mpps', 'bpps',
'bandwidth', 'capacity', 'power', 'strength',
'layer 4', 'layer 7', 'l4', 'l7', 'amplification',
'reflection', 'bypass', 'udp flood', 'syn flood',
'http flood', 'rate limit']
```

The last list is used for commercialisation purposes. It matches messages that discuss ways to buy or sell attacks, stressers and booters, and test attacks.

```
['stresser', 'booter', 'panel', 'tier', 'plans',
'pricing', 'subscription', 'reseller', 'api access',
'uptime guarantee', 'sla', 'test attack',
'test hit', 'demo']
```

A.4 Detailed Power Proof results



Figure 14: A Telegram message from a bot with a detailed report showing IP protocols, source ports, sources ASes and source countries.